OpenConjecture — Data Dictionary

Hugging Face dataset davisrbr/openconjecture · Mathematical conjectures auto-extracted from arXiv LaTeX sources, with LLM open-status labels and interestingness/viability scores

CONFIGS	CONJECTURES	PAPERS SCANNED	SPLITS	ASSESSOR
2	**4,147**	**25,395**	**train**	**gpt-5-mini**
conjectures · papers	3,335 labeled real & open	arXiv, 2017–2026	only split, both configs	gpt5mini-v5-open-exact-v1

Provenance model: each conjecture row records exactly where it was extracted from (`source_file`, `start_line`–`end_line` inside the arXiv e-print at `source_url`), plus a `content_hash`. The two configs join on `arxiv_id`.

> **Licensing caveat:** 2,181 of 4,147 conjecture rows (53%) have `text_withheld = true` — `body_tex` and `plain_text` are empty because the paper is under arXiv's non-exclusive-distribution license, which does not permit republication. Text is included only for CC-licensed papers. Withheld text can be reconstructed by downloading `source_url` and reading the recorded line span.

Config 1: `conjectures` — 4,147 rows × 41 fields · one row per extracted conjecture

abs_url	string	URL of the paper's arXiv abstract page.	length (chars): min 32 · max 34 · mean 33.12	`http://arxiv.org/abs/2607.02500v1`
arxiv_id	string	arXiv identifier of the source paper, including version suffix (e.g. `2607.02500v1`). Join key between the two configs.	length (chars): min 10 · max 12 · mean 11.89	`2607.02500v1`
authors	list[string]	List of author names, in paper order.	length: min 1 · max 44 · mean 2.44	`['Lawrence Hollom']`
body_tex	string	Raw LaTeX body of the conjecture environment. **Empty string when `text_withheld` is true.**	length (chars): min 0 · max 5,785 · mean 145.93	`\label{conj:ak} If a poset $P$ contains no infinite antichain then there exists a chain $C$ and a partitio…`
categories	list[string]	All arXiv subject categories assigned to the paper (e.g. `math.CO`, `math.PR`).	length: min 0 · max 5 · mean 1.4	`['math.CO']`

comments	string	Author-supplied arXiv comments field (page counts, revision notes, etc.).	744 distinct values top: `(empty)` 2,043 · `19 pages` 53 · `14 pages` 40	`51 pages, 2 figures. The content here used to be part of 2411.16844 (which now contains just the counterexampl…`
content_hash	string	SHA-256 hash (hex) of the conjecture content; stable identifier even when the text itself is withheld.	length (chars): min 64 · max 64 · mean 64	`882f0577a1dbd2bc0fe8fb066127c34db35548dfe9da65ac2421058319283707`
doi	string	Digital Object Identifier of the published version, when available. Empty for most records.	18 distinct values top: `(empty)` 4,109 · `10.1007/s11787-025-00394-2` 7 · `10.56994/JXM.001.002.003` 6	*(empty in sample)*
end_line	int64	Last line (1-based) of the conjecture environment in `source_file`.	min 19 · max 24,374 · mean 1,028.57 · median 503	`674`
id	int64	Unique integer identifier for the conjecture record.	min 1 · max 4,613 · mean 2,219 · median 2,119	`4411`
index_in_file	int64	1-based ordinal of this conjecture among the conjecture environments found in `source_file`.	min 1 · max 36 · mean 2.41 · median 1	`1`
journal_ref	string	Journal reference string supplied by the authors, when available. Empty for most records.	17 distinct values top: `(empty)` 4,113 · `Logica Universalis 19, 799–840 (2025)` 7 · `Journal of Experimental Mathematics 1 (2025) 239––256` 6	*(empty in sample)*
latest_assessment_version	string	Version tag of the assessment prompt/pipeline (constant: `gpt5mini-v5-open-exact-v1`).	1 distinct value top: `gpt5mini-v5-open-exact-v1` 4,147	`gpt5mini-v5-open-exact-v1`
latest_evidence_snippet	string	Quotation from the paper the model cited as evidence for its label.	length (chars): min 0 · max 337 · mean 74.12	`This conjecture is now known to be false: the author [hollom2024resolution] constructed a counterexample, ... …`

latest_interestingness_confidence	float64	Model confidence in the interestingness score, 0–1.	min 0 · max 0.95 · mean 0.56 · median 0.68	`0.0`
latest_interestingness_rationale	string	Free-text rationale for the interestingness score.	length (chars): min 0 · max 296 · mean 77.33	*(empty in sample)*
latest_interestingness_score	float64	LLM-judged mathematical interestingness of the conjecture, 0–1. Zeroed for records labeled `not_real_conjecture`.	min 0 · max 0.97 · mean 0.51 · median 0.58	`0.0`
latest_label	string	LLM classification of the extracted statement: `real_open_conjecture`, `not_real_conjecture` (e.g. already resolved, or not actually a conjecture), or `uncertain`.	3 distinct values top: `real_open_conjecture` 3,335 · `not_real_conjecture` 786 · `uncertain` 26	`not_real_conjecture`
latest_label_confidence	float64	Model confidence in `latest_label`, 0–1.	min 0 · max 1 · mean 0.89 · median 0.9	`0.98`
latest_label_model	string	LLM used to produce the most recent assessment (constant: `gpt-5-mini`).	1 distinct value top: `gpt-5-mini` 4,147	`gpt-5-mini`
latest_label_rationale	string	Free-text rationale the model gave for its label.	length (chars): min 0 · max 263 · mean 73.78	`The paper explicitly states the Aharoni--Korman (fishbone) conjecture is now known to be false and cites the a…`
latest_labeled_at	timestamp[s]	Timestamp when the most recent LLM assessment was produced.	range 2026-03-06 → 2026-07-05	`2026-07-05T14:43:33`
latest_viability_confidence	float64	Model confidence in the viability score, 0–1.	min 0 · max 0.9 · mean 0.34 · median 0.35	`0.0`
latest_viability_rationale	string	Free-text rationale for the viability score.	length (chars): min 0 · max 288 · mean 74.07	*(empty in sample)*
latest_viability_score	float64	LLM-judged viability — how tractable/attackable the conjecture appears — 0–1. Low median (0.25) reflects that most open conjectures are judged hard.	min 0 · max 0.9 · mean 0.28 · median 0.25	`0.0`
license_family	string	License grouped into a family: `arxiv_nonexclusive_distrib`, `cc_by`, `cc_by_nc_nd`, `cc_by_nc_sa`, `cc_by_sa`, or `missing`.	6 distinct values top: `arxiv_nonexclusive_distrib` 2,181 · `cc_by` 1,546 · `cc_by_nc_nd` 186	`cc_by`

`license_url`	`string`	License URL declared for the paper on arXiv, verbatim.	6 distinct values top: `http://arxiv.org/licenses/nonexclusive-distrib/1.0/` 2,181 · `http://creativecommons.org/licenses/by/4.0/` 1,546 · `http://creativecommons.org/licenses/by-nc-nd/4.0/` 186	`http://creativecommons.org/licenses/by/4.0/`
`normalized_license_url`	`string`	License URL normalized to a canonical HTTPS form.	6 distinct values top: `https://arxiv.org/licenses/nonexclusive-distrib/1.0/` 2,181 · `https://creativecommons.org/licenses/by/4.0/` 1,546 · `https://creativecommons.org/licenses/by-nc-nd/4.0/` 186	`https://creativecommons.org/licenses/by/4.0/`
`pdf_url`	`string`	URL of the paper's PDF on arXiv.	length (chars): min 32 · max 34 · mean 33.54	`https://arxiv.org/pdf/2607.02500v1`
`plain_text`	`string`	Plain-text rendering of the conjecture (inline math kept as LaTeX). **Empty string when `text_withheld` is true.**	length (chars): min 0 · max 3,338 · mean 100.92	`If a poset $P$ contains no infinite antichain then there exists a chain $C$ and a partition of $P$ into disjoi…`
`primary_category`	`string`	The paper's primary arXiv subject category. May be empty for some records.	62 distinct values top: `math.CO` 983 · `(empty)` 766 · `math.NT` 550	`math.CO`
`publication_decision`	`string`	Pipeline decision on whether the conjecture/paper text may be republished: `publish_text` or `withhold_text`.	2 distinct values top: `withhold_text` 2,181 · `publish_text` 1,966	`publish_text`
`publication_policy_version`	`string`	Version tag of the publication/licensing policy applied (constant: `hf-publication-v2`).	1 distinct value top: `hf-publication-v2` 4,147	`hf-publication-v2`
`publication_text_allowed`	`bool`	Boolean form of `publication_decision`: true when the text may be included in the dataset.	top: `False` 2,181 · `True` 1,966	`True`

publication_text_reason	string	Policy rule that produced the publication decision (e.g. CC license treated as publishable; arXiv non-exclusive-distribution license requires withholding).	4 distinct values top: `arxiv_nonexclusive_distribution_license` 2,181 · `creativecommons_license_treated_as_publishable` 1,581 · `creativecommons_noncommercial_license_allowed_for_noncommercial_release` 325	`creativecommons_license_treated_as_publishable`
published_at	timestamp[s]	Timestamp when the paper version was first published on arXiv.	range 2020-11-02 → 2026-07-02	`2026-07-02T17:58:25`
source_file	string	Name of the `.tex` file inside the e-print source archive from which the conjecture was extracted.	length (chars): min 5 · max 129 · mean 17.2	`main.tex`
source_url	string	URL of the paper's LaTeX e-print source archive on arXiv — use this to retrieve withheld conjecture text.	length (chars): min 36 · max 38 · mean 37.89	`https://arxiv.org/e-print/2607.02500v1`
start_line	int64	First line (1-based) of the conjecture environment in `source_file`.	min 17 · max 24,372 · mean 1,023.21 · median 500	`671`
text_withheld	bool	True when conjecture text (`body_tex` , `plain_text`) is withheld for licensing reasons — 53% of rows. Metadata and source line references remain available.	top: `True` 2,181 · `False` 1,966	`False`
title	string	Title of the source arXiv paper.	length (chars): min 11 · max 166 · mean 63.64	`The structure of FAC posets and the Aharoni–Korman conjecture`
updated_at	timestamp[s]	Timestamp of the most recent update to the paper on arXiv.	range 2020-11-02 → 2026-07-02	`2026-07-02T17:58:25`

Config 2: `papers` — 25,395 rows × 25 fields · one row per scanned arXiv paper

abs_url	string	URL of the paper's arXiv abstract page.	length (chars): min 32 · max 35 · mean 33.13	`http://arxiv.org/abs/2607.02511v1`
arxiv_id	string	arXiv identifier of the source paper, including version suffix (e.g. `2607.02500v1`). Join key between the two configs.	length (chars): min 10 · max 13 · mean 11.9	`2607.02511v1`

authors	list[string]	List of author names, in paper order.	length: min 1 · max 48 · mean 2.37	`['Zakaria Derbazi']`
categories	list[string]	All arXiv subject categories assigned to the paper (e.g. `math.CO`, `math.PR`).	length: min 0 · max 7 · mean 1.45	`['math.CA', 'math.PR']`
comments	string	Author-supplied arXiv comments field (page counts, revision notes, etc.).	length (chars): min 0 · max 432 · mean 12.77	`19 pages`
conjecture_count	int64	Number of conjecture environments extracted from this paper (0 for most papers).	min 0 · max 36 · mean 0.16 · median 0	`0`
conjectures_with_public_text	int64	How many of this paper's conjectures have their text published in the dataset.	min 0 · max 27 · mean 0.08 · median 0	`0`
conjectures_withheld_text	int64	How many of this paper's conjectures have their text withheld for licensing reasons.	min 0 · max 36 · mean 0.09 · median 0	`0`
doi	string	Digital Object Identifier of the published version, when available. Empty for most records.	280 distinct values top: `(empty)` 25,116 · `10.2422/2036-2145.202505_012` 1 · `10.1007/s11040-020-09371-1` 1	*(empty in sample)*
ingested_at	timestamp[s]	Timestamp when the paper was ingested by the OpenConjecture pipeline.	range 2026-03-03 → 2026-07-05	`2026-07-05T14:03:58`
journal_ref	string	Journal reference string supplied by the authors, when available. Empty for most records.	270 distinct values top: `(empty)` 25,123 · `The 23rd IFAC World Congress, IFAC, Aug 2026, Busan, South Korea` 2 · `Journal of Applied Mathematics, Statistics and Informatics (JAMSI) ,JAMSI, 20 (2024), No. 1` 2	*(empty in sample)*
license_family	string	License grouped into a family: `arxiv_nonexclusive_distrib`, `cc_by`, `cc_by_nc_nd`, `cc_by_nc_sa`, `cc_by_sa`, or `missing`.	6 distinct values top: `arxiv_nonexclusive_distrib` 13,256 · `cc_by` 9,362 · `cc_by_nc_nd` 1,231	`cc_by`

license_url	string	License URL declared for the paper on arXiv, verbatim.	6 distinct values top: `http://arxiv.org/licenses/nonexclusive-distrib/1.0/` 13,256 · `http://creativecommons.org/licenses/by/4.0/` 9,362 · `http://creativecommons.org/licenses/by-nc-nd/4.0/` 1,231	`http://creativecommons.org/licenses/by/4.0/`
normalized_license_url	string	License URL normalized to a canonical HTTPS form.	6 distinct values top: `https://arxiv.org/licenses/nonexclusive-distrib/1.0/` 13,256 · `https://creativecommons.org/licenses/by/4.0/` 9,362 · `https://creativecommons.org/licenses/by-nc-nd/4.0/` 1,231	`https://creativecommons.org/licenses/by/4.0/`
pdf_url	string	URL of the paper's PDF on arXiv.	length (chars): min 32 · max 35 · mean 33.53	`https://arxiv.org/pdf/2607.02511v1`
primary_category	string	The paper's primary arXiv subject category. May be empty for some records.	135 distinct values top: `(empty)` 4,676 · `math.CO` 1,967 · `math.AP` 1,909	`math.CA`
publication_decision	string	Pipeline decision on whether the conjecture/paper text may be republished: `publish_text` or `withhold_text`.	2 distinct values top: `withhold_text` 13,256 · `publish_text` 12,139	`publish_text`
publication_policy_version	string	Version tag of the publication/licensing policy applied (constant: `hf-publication-v2`).	1 distinct value top: `hf-publication-v2` 25,395	`hf-publication-v2`
publication_text_allowed	bool	Boolean form of `publication_decision`: true when the text may be included in the dataset.	top: `False` 13,256 · `True` 12,139	`True`
publication_text_reason	string	Policy rule that produced the publication decision (e.g. CC license treated as publishable; arXiv non-exclusive-distribution license requires withholding).	4 distinct values top: `arxiv_nonexclusive_distribution_license` 13,256 · `creativecommons_license_treated_as_publishable` 9,672 · `creativecommons_noncommercial_license_allowed_for_noncommercial_release` 1,903	`creativecommons_license_treated_as_publishable`
published_at	timestamp[s]	Timestamp when the paper version was first published on arXiv.	range 2017-02-02 → 2026-07-02	`2026-07-02T17:59:46`

source_url	string	URL of the paper's LaTeX e-print source archive on arXiv — use this to retrieve withheld conjecture text.	length (chars): min 36 · max 39 · mean 37.9	`https://arxiv.org/e-print/2607.02511v1`
summary	string	Paper abstract as provided by arXiv.	length (chars): min 43 · max 3,020 · mean 942.2	`This paper develops a probabilistic sign rule for quotients of functions represented by positive series or int…`
title	string	Title of the source arXiv paper.	length (chars): min 7 · max 225 · mean 73.55	`A Probabilistic Sign Rule for Quotients of Positive Series and Integral Transforms`
updated_at	timestamp[s]	Timestamp of the most recent update to the paper on arXiv.	range 2017-02-02 → 2026-07-02	`2026-07-02T17:59:46`

Generated 2026-07-10 from the Hugging Face datasets-server API (`/info`, `/statistics`, `/first-rows`). Statistics computed over the full train split of each config. Example values taken from the first row of each config.